

Problem Statement ID : PS 2

- Web Forum Threat Intelligence Scraper & Analyzer
- Team Name :Hack Horizon
- TEAM ID : HK - 112
- TEAM MEMBERS : Vishal Jangir, Ravi, Ashish, Kunal

Problem & SOLUTION

PROBLEM: Rising Cyber Threats in India

- Increasing attacks on critical infrastructure
- Credentials leaked daily on public forums & Telegram
- Planned attack discussions (ransomware, DDoS, exploits)
- Insider threats often go undetected
- Manual monitoring is slow

Impact: Data breaches, financial loss, national security risks

Flow of Solution

Web Forums / Telegram → Scraper Engine
Employee Logs → Log Parser

Processing Layer:

- NLP analysis
- ML anomaly detection
- Regex & keyword rules

Output:

Threat scoring → Secure DB → Dashboard → Alerts

TECH STACK & APPROACH

- Python 3
- BeautifulSoup / Scrapy / Selenium
- spaCy & HuggingFace
- Scikit-learn
- PostgreSQL / MongoDB
- Flask / FastAPI
- React / Streamlit

UNIQUENESS & INNOVATION FACTOR

- Working forum scraper
- API leak detection
- Insider anomaly simulation
- Dashboard prototype

Demo Includes:

- Scraped post → detected leak
- Mock log → flagged anomaly

Challenges, Feasibility & Impact

Challenges:

- Anti-bot protection
- False positives
- Ethical boundaries

Feasibility:

- Open-source tools
- Cloud deployable
- Scalable design

Impact:

- Early warning for organizations
- Helps prevent credential & insider breaches



External:

- Regex detection of credentials
- Keyword detection
- NLP context analysis

Internal:

- Login pattern changes
- Unusual file access
- USB anomalies

Scoring:

Severity × Credibility × Urgency × Sector Impact



HACK KRMU
5.0

THANK YOU